ModelIt!

Teacher's Guide

How Do Stars Forge Every Atom in Your Body?

Modeling Stellar Nucleosynthesis and the Origin of the Elements

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-ESS1-3, HS-PS1-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS1-3, HS-PS1-5

Communicate scientific ideas about the way stars, over their life cycle, produce elements. Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model of stellar nucleosynthesis that connects nuclear physics to stellar evolution and element production across cosmic timescales.
Disciplinary Core Idea	**ESS1.A The Universe and Its Stars / PS1.C Nuclear Processes** — Stars produce elements through nuclear fusion in their cores, with the specific elements produced determined by stellar mass and lifecycle stage.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students trace matter transformation from primordial hydrogen through stellar nucleosynthesis to the heavy elements found in planets and living organisms.

CAST Testing Alignment

MULTIPLE CHOICE:

A star 25 times the mass of our Sun has built up an iron core after fusing successively heavier elements. What happens next, according to your model?

CONSTRUCTED RESPONSE:

Using your model, explain why the calcium in your bones and the iron in your blood required different types of stars to produce. Reference the role of Gravitational Pressure and Stellar Core Temperature in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

You are literally made of stardust — and that is not a metaphor. The calcium in your bones was forged in a supergiant star. The iron in your blood was the last element produced before a star collapsed and exploded. The oxygen you're breathing right now was fused from carbon in a stellar core at 600 million degrees. Understanding stellar nucleosynthesis is understanding the origin of everything.

The Cosmic Origin of Elements

After the Big Bang, the universe contained only hydrogen (~75%), helium (~25%), and trace amounts of lithium. Every other element on the periodic table was created later inside stars. This process — stellar nucleosynthesis — occurs when gravitational compression heats stellar cores to temperatures where atomic nuclei can overcome their electromagnetic repulsion and fuse. The energy released by fusion is what makes stars shine.

The Fusion Sequence

Stars fuse elements in a specific order dictated by the energy required to overcome nuclear repulsion. Hydrogen fuses to helium at 10 million K (our Sun does this). Helium fuses to carbon at 100 million K. Carbon to neon at 600 million K. Neon to oxygen at 1.2 billion K. Oxygen to silicon at 1.5 billion K. Silicon to iron at 2.7 billion K. Each stage requires higher temperatures and produces less energy — a star that burns hydrogen for 10 million years may burn silicon for only one day.

The Iron Catastrophe

Iron-56 has the highest binding energy per nucleon of any element. This means fusing iron into heavier elements does not release energy — it requires energy input. When a massive star accumulates an iron core of about 1.4 solar masses (the Chandrasekhar limit), fusion stops. In less than a second, the core collapses from the size of Earth to a sphere 20 km across. The resulting shockwave — a supernova — is briefly brighter than an entire galaxy and creates all elements heavier than iron through rapid neutron capture (the r-process).

From Stardust to You

The elements scattered by supernovae mix with interstellar gas and eventually collapse into new star systems — complete with planets. Our solar system formed 4.6 billion years ago from a cloud enriched by multiple generations of stellar deaths. The Earth's composition reflects this cosmic heritage: an iron core, a silicate mantle, a nitrogen-oxygen atmosphere, and carbon-based life. Every atom in your body has been inside at least one star, and the heaviest atoms required the most violent events in the universe to create.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of stellar nucleosynthesis: Stellar Core Temperature, Hydrogen Fusion Rate, Helium Accumulation, Gravitational Pressure, Radiation Output, Element Production, and Star Lifecycle Stage — with Gravitational Pressure as the external driver.

E **Establish Relationships**

Students determine that Gravitational Pressure drives Stellar Core Temperature, which enables Hydrogen Fusion Rate. Fusion produces Radiation Output (opposing gravity) and Helium Accumulation (fuel for the next stage). Element Production advances as each fuel is exhausted.

V **Visualize & Model**

Students build a computational model showing the balance between gravitational collapse and radiation pressure, and how fuel exhaustion drives transitions between lifecycle stages.

E **Evaluate Outcomes**

Students run sun-like, massive star, and red dwarf scenarios to compare element production capacity, lifecycle duration, and end states across different stellar masses.

R **Revise & Extend**

Students add Stellar Mass, Neutrino Emission, or Metallicity to explore how initial conditions and energy loss mechanisms affect stellar evolution pathways.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Hold up your hand. The iron in your blood was forged in a star that exploded before our Sun existed. The calcium in your bones came from a different star. You are made of dead stars."*

Watch For

Let the statement land. Show the periodic table with elements color-coded by their stellar origin. Give students 30 seconds to process.

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling the most powerful factories in the universe — stellar cores where atoms are literally built from scratch."*

Watch For

Have students read objectives. Pre-teach 'nucleosynthesis' and 'binding energy.' Quick-write: What do you think powers a star?

Slide 3: Big Question

3 min

Say: *"The Big Bang made hydrogen and helium. That's it. So where did the other 90+ elements come from?"*

Watch For

Students examine a periodic table and identify which elements are in the human body. Challenge: Where did each one originate?

Slide 4: LEVER Framework

2 min

Say: *"We're going to model the entire life of a star — from its birth burning hydrogen to its death creating the heaviest elements."*

Watch For

Preview each LEVER step. Emphasize that stellar mass is the master variable that determines everything.

Slide 5: Activity 1: Components

8 min

Say: *"A star is a battleground between two forces: gravity trying to crush it and radiation trying to blow it apart. Everything else follows from that battle."*

Watch For

Guide component sorting. Discuss why Gravitational Pressure is external (set by the star's mass at birth) while other components respond to it.

Slide 6: Activity 2: Connections

8 min

Say: *"When hydrogen runs out, what happens to the balance between gravity and radiation? That moment determines everything."*

Watch For

Help students trace the feedback loop: gravity compresses, temperature rises, fusion ignites, radiation pushes back. Identify what breaks this balance.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's watch three different stars live and die. A star like our Sun. A monster 20 times larger. And a tiny red dwarf."*

Watch For

Students predict element production and lifespan for each mass. Run simulations. Compare the dramatically different outcomes.

Slide 8: Discoveries

5 min

Say: *"Your body is a periodic table of stellar debris. Different elements required different stars, different temperatures, and different deaths."*

Watch For

Create a class 'element origin map' connecting body elements to their stellar sources. Discuss why supernovae are essential for life.

Slide 9: STEM Challenge

12 min

Say: *"A science museum wants visitors to truly understand that they are made of stars. Design the exhibit that makes it real."*

Watch For

Groups design interactive stellar evolution visualizations. Must show mass-dependent outcomes and connect to human body composition.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Gravitational Pressure
- **INTERNAL:** Stellar Core Temperature, Hydrogen Fusion Rate, Helium Accumulation, Radiation Output, Element Production, Star Lifecycle Stage

Relationships

- **Gravitational Pressure to Stellar Core Temperature:** POSITIVE (+) — Greater Gravitational Pressure compresses the core more intensely, converting gravitational potential energy to thermal energy and raising core temperature to the thresholds required for each successive fusion stage.
- **Stellar Core Temperature to Hydrogen Fusion Rate:** POSITIVE (+) — Higher core temperature dramatically increases the rate of hydrogen fusion because nuclear fusion probability depends exponentially on temperature — even small temperature increases produce large increases in reaction rates through quantum tunneling.
- **Hydrogen Fusion Rate to Radiation Output:** POSITIVE (+) — Faster hydrogen fusion releases more energy per second as gamma rays and neutrinos, increasing the outward Radiation Output that supports the star against gravitational collapse.

Key Discoveries

- Stars are element factories — the balance between Gravitational Pressure and Radiation Output creates the conditions for nuclear fusion to forge progressively heavier elements
- Each fusion stage requires higher temperatures and produces heavier elements, but releases less energy — creating an inevitable countdown to fuel exhaustion
- Iron is the endpoint of energy-releasing fusion because its nucleus has the highest binding energy per nucleon — fusing iron absorbs energy instead of releasing it, triggering core collapse
- The elements in your body required at least two generations of stars — hydrogen and helium from the Big Bang, carbon and oxygen from medium stars, and iron and heavier elements from supernovae

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Stars burn like fires"	Stars do not undergo chemical combustion (burning). They produce energy through nuclear fusion — combining atomic nuclei at temperatures of millions of degrees. Chemical burning involves electron interactions and produces thousands of degrees; nuclear fusion involves nuclear forces and produces millions to billions of degrees. The energy output per kilogram of fuel is roughly 10 million times greater for fusion than for combustion.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The Sun will explode as a supernova"	Only stars roughly 8 or more times the mass of the Sun have enough gravitational pressure to progress through all fusion stages to iron core collapse. Our Sun will exhaust its hydrogen, swell into a red giant, fuse helium into carbon and oxygen, then gently shed its outer layers as a planetary nebula — leaving behind a white dwarf the size of Earth. No explosion, no supernova.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heavy elements are made during normal stellar fusion"	Normal stellar fusion can only produce elements up to iron (atomic number 26). All elements heavier than iron — including copper, silver, gold, uranium — require the extreme neutron flux of a supernova explosion or neutron star merger. These events produce enormous numbers of free neutrons that are rapidly captured by nuclei, building up to the heaviest elements in seconds.